



Fitness and Diet Tracker with Personalized Recommendations

SCENARIO-BASED REQUIREMENT ANALYSIS

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ABSTRACT

The growing global emphasis on preventive healthcare and healthy lifestyles has driven demand for digital tools that help individuals monitor and improve their physical wellbeing. Conventional fitness applications largely function as simple activity loggers, offering step counts or calorie totals without truly personalized guidance, leaving users to interpret their own data and design their own workout and diet plans. The Fitness and Diet Tracker with Personalized Recommendations is proposed as an AI-enabled platform that collects user health data — including age, weight, height, BMI, and fitness goals — and continuously monitors daily physical activity and nutritional intake to generate customized workout schedules and meal plans tailored to each user's preferences and health conditions. The platform synchronizes automatically with wearable devices to capture activity, heart-rate, and sleep metrics, visualizes progress through interactive dashboards and reports, and sends timely reminders for workouts, hydration, and meals, all while safeguarding the privacy and security of personal health information.

The system is designed as a modular, cloud-hosted platform capable of scaling from a small group of early adopters to a large user base spanning individuals, fitness trainers, and nutritionists. Its layered architecture separates data collection (activity and nutrition logging, wearable synchronization), intelligence (AI-driven recommendation and progress-analysis engines), and presentation (dashboards, mobile app, and notifications), allowing each layer to evolve independently as wearable technology, recommendation algorithms, and nutrition databases mature. By unifying activity tracking, nutrition logging, personalized recommendations, and progress visualization into a single platform, the system aims to improve user adherence to fitness and nutrition goals, encourage consistent healthy-lifestyle habits, and support better long-term health outcomes. In doing so, it positions itself not merely as an activity logger but as a decision-support companion that helps users move from unguided, sporadic effort toward informed, personalized, and sustained wellness management.

Keywords: Fitness Tracking, Personalized Diet Recommendation, Health Monitoring

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1. INTRODUCTION

1.1 Purpose

This document specifies the software requirements for the Fitness and Diet Tracker with Personalized Recommendations, an AI-enabled platform for monitoring users' physical activity and nutritional habits and generating individualized workout and diet guidance. It defines what the system must do (functional requirements) and how well it must perform (non-functional requirements), and serves as the baseline reference for design, development, testing, and evaluation.

1.2 Business Domain

The system operates in the digital health and wellness domain, at the intersection of consumer wearable technology, AI/ML-based personalization, and cloud-based health-data platforms. It targets individual users, fitness trainers, and nutritionists who need to convert raw activity and dietary data into actionable, personalized guidance that supports consistent, healthy lifestyle habits.

1.3 Intended Audience

This SRS is intended for the software development team, project evaluators, fitness and nutrition stakeholders, quality assurance testers, and future maintainers of the Fitness and Diet Tracker platform.

1.4 Definitions and Abbreviations

Term	Meaning
BMI	Body Mass Index
BMR	Basal Metabolic Rate — the rate at which the body burns calories at rest
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
SRS	Software Requirement Specification
IoT	Internet of Things (used for wearable device connectivity)
KPI	Key Performance Indicator
SLA	Service Level Agreement

2. Problem Description

Individuals seeking to improve their health currently rely on fragmented tools — basic step counters, generic workout apps, or manual food diaries — that log activity and intake but provide little to no personalized guidance. These tools lack the intelligence to translate a user's age, weight,

BMI, preferences, and health conditions into a coherent, individualized workout schedule and diet plan, and they rarely account for changing progress over time. As a result, users struggle with inconsistent routines, poorly matched diet plans, missed workouts, and low adherence to fitness goals, ultimately leading to poor long-term health outcomes. Existing applications also lack comprehensive progress tracking and often fail to integrate seamlessly with wearable devices, leaving users to manually reconcile data from multiple sources. The Fitness and Diet Tracker with Personalized Recommendations addresses this gap by combining user health profiles, wearable-device telemetry, and AI-driven recommendation models into a single platform that gives users continuous, personalized, and actionable guidance instead of raw activity logs alone — shifting fitness and nutrition management from unguided self-tracking to informed, data-driven, and consistent lifestyle improvement.

3. Scope of the System

3.1 In-Scope Features

- Collection of user health profile data, including age, weight, height, BMI, and fitness goals
- Daily tracking of physical activity (steps, calories burned, workout duration) and nutritional intake (meals, calories, macronutrients)
- AI-based generation of personalized workout schedules based on user profile, goals, and progress
- AI-based generation of personalized diet/meal plans based on preferences and health conditions
- Automatic synchronization of health metrics from wearable devices (steps, heart rate, sleep)
- Interactive progress dashboards and visual reports tracking weight, activity, and nutrition trends
- Analytical and progress reporting with export capability
- Reminder and notification system for workouts, hydration, and meals
- User and role management for users, trainers/nutritionists, and administrators
- Secure storage and management of personal health information

3.2 Out-of-Scope Features

- Manufacturing or hardware production of wearable devices or fitness equipment
- Telemedicine or direct doctor/physician consultation booking
- E-commerce marketplace for supplements, equipment, or meal-kit delivery
- Social media / community feed features (public sharing, following other users)
- Insurance claims or wellness-incentive settlement integration
- Grocery ordering or delivery integration

4. STAKEHOLDERS AND BUSINESS CONTEXT

Stakeholder	Role / Interest in the System
User / Fitness Enthusiast	Logs activity and meals, receives personalized workout and diet recommendations, tracks progress
Nutritionist / Fitness Trainer	Reviews AI-generated plans, provides professional guidance, adjusts recommendations for assigned users
System Administrator	Manages user accounts, roles, permissions, and system configuration
Wearable Device Provider (external)	Provides real-time activity, heart-rate, and sleep data via API integration
Business/Platform Owner	Grows user engagement and retention, supports subscription/wellness revenue goals
Regulatory Body (indirect)	Concerned with personal health data privacy and protection compliance

Business Context: Users are under personal motivation to achieve fitness and health goals but are frequently held back by generic, one-size-fits-all guidance. The Fitness and Diet Tracker positions itself as the analytical and recommendation layer that connects an individual's activity and nutrition data to sustained, personalized lifestyle decisions.

5. FUNCTIONAL REQUIREMENTS

ID	Requirement
FR1	The system shall collect user health profile data, including age, weight, height, BMI, and fitness goals.
FR2	The system shall track daily physical activity, including steps, calories burned, and workout duration.
FR3	The system shall track daily nutritional intake, including meals, calories consumed, and macronutrient breakdown.
FR4	The system shall use an AI/ML-based engine to generate personalized workout schedules based on user profile and goals.
FR5	The system shall use an AI/ML-based engine to generate personalized diet and meal plans based on user preferences and health conditions.
FR6	The system shall automatically synchronize activity, heart-rate, and sleep data from connected wearable devices.
FR7	The system shall provide an interactive dashboard displaying activity, nutrition, and progress KPIs to the user.

ID	Requirement
FR8	The system shall generate analytical progress reports on activity, nutrition, and goal attainment over selectable time periods.
FR9	The system shall send real-time reminders and notifications for scheduled workouts, hydration, and meal times.
FR10	The system shall allow users to set, edit, and track fitness goals such as weight loss, muscle gain, or maintenance.
FR11	The system shall allow administrators to manage user accounts and role-based access permissions.
FR12	The system shall log historical activity and nutrition data with timestamps for trend analysis and auditing.
FR13	The system shall allow users to view, edit, or manually log meals and exercises not captured automatically.
FR14	The system shall integrate with external wearable-device APIs to retrieve real-time health and activity metrics.
FR15	The system shall support data export (CSV/PDF) of progress reports for offline review.
FR16	The system shall allow users to manually override or adjust AI-generated workout and diet recommendations.
FR17	The system shall maintain an audit log of recommendation changes and plan adjustments made by users or trainers.
FR18	The system shall trigger an alert when a user misses activity or calorie targets for a configured period.
FR19	The system shall allow administrators to configure recommendation and alert thresholds (e.g., daily step goal, calorie limits).
FR20	The system shall provide a historical trend view of a user's weight, BMI, and activity levels for long-term progress tracking.

6. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	The system shall reflect logged activity and nutrition updates on the dashboard within 5 seconds. Personalized recommendations shall be generated within 3 seconds of a request.

Category	Requirement
Scalability	The system shall support at least 10,000 concurrent users without degradation in dashboard responsiveness, and shall scale horizontally as the user base grows.
Reliability	The platform shall maintain 99.5% uptime, with automatic failover for critical tracking and reminder services.
Security	All data transmission between wearable devices, the mobile app, and the cloud platform shall be encrypted (TLS 1.2+). Role-based access control shall restrict health-data visibility by user type.
Usability	The mobile and web dashboard shall be usable by non-technical users with no more than 15 minutes of onboarding, following a clean, intuitive UI.
Maintainability	The system shall follow a modular microservice architecture allowing individual components (recommendation engine, tracking engine, dashboard) to be updated independently without full-system redeployment.
Availability	Core tracking, logging, and reminder services shall be available 24/7, including during scheduled maintenance of non-critical modules.
Interoperability	The system shall expose REST APIs to integrate with third-party wearable-device platforms (e.g., Fitbit, Google Fit, Apple Health) and nutrition databases.
Portability	The web dashboard shall function correctly across major browsers (Chrome, Edge, Firefox, Safari), and the mobile app shall support both Android and iOS.
Data Retention	Historical activity, nutrition, and progress data shall be retained for a minimum of 2 years for trend analysis and compliance.
Accuracy	The recommendation model shall maintain a minimum accuracy threshold of 85% against validated historical outcomes before being used for production recommendations.

7. ACTORS AND USE CASES

7.1 Primary and Secondary Actors

Primary Actors:

- User / Fitness Enthusiast – Logs activity and meals, sets goals, views dashboards/reports, receives recommendations
- Nutritionist / Fitness Trainer – Reviews and adjusts AI-generated plans for assigned users
- System Administrator – Manages user accounts, roles, and system configuration

Secondary Actors:

- Wearable Device System – Supplies real-time activity, heart-rate, and sleep data
- AI Recommendation Engine – Runs workout and diet personalization models
- Notification Gateway – Delivers reminders and alerts via app/SMS/email

7.2 Use Case Descriptions

UC-1: Track Daily Activity

- **Actors:** User; Wearable Device System (trigger)
- **Precondition:** User's wearable device or mobile app is active and connected.
- **Main Flow:** (1) Wearable device or manual entry captures steps, calories burned, and workout duration. (2) Data is transmitted to the cloud backend. (3) System updates the user's daily activity log. (4) User views updated activity status on the dashboard.
- **Alternate Flow:** If wearable connectivity is lost, system flags the device as “offline” and prompts the user for manual entry.
- **Postcondition:** Daily activity data is current and visible in real time.

UC-2: Generate Personalized Workout Plan

- **Actors:** AI Recommendation Engine; User
- **Precondition:** Sufficient user profile and activity history exists.
- **Main Flow:** (1) Recommendation engine analyzes user profile, goals, and recent activity. (2) Model computes a suitable workout schedule. (3) Plan is generated and presented to the user. (4) User accepts or requests adjustments.
- **Postcondition:** A personalized workout plan is available and visible to the user.

UC-3: Generate Personalized Diet Plan

- **Actors:** AI Recommendation Engine; User
- **Precondition:** User profile includes dietary preferences and health conditions.
- **Main Flow:** (1) System checks current nutritional intake against the user's goal-based targets. (2) System applies preferences and health-condition constraints. (3) System computes a suitable meal plan. (4) Recommendation is sent to the user for approval or auto-applied per configuration.
- **Alternate Flow:** If a preference conflicts with a health condition (e.g., an allergy), system excludes the conflicting food items and flags the substitution.
- **Postcondition:** A diet plan is generated and communicated to the user.

UC-4: Sync Wearable Device Data

- **Actors:** User; Wearable Device System
- **Precondition:** User has linked a supported wearable device account.
- **Main Flow:** (1) User initiates or schedules a sync. (2) System retrieves activity, heart-rate, and sleep data via the device API. (3) Data is merged into the user's activity log. (4) Dashboard is updated with the latest metrics.
- **Postcondition:** Wearable data is reflected in the user's activity history.

UC-5: View Progress Dashboard

- **Actors:** User
- **Main Flow:** User logs in; system aggregates recent activity, nutrition, and goal-progress data, and renders them on a consolidated dashboard view.

UC-6: Generate Progress Reports

- **Actors:** User
- **Main Flow:** User selects a date range and report type; system compiles activity, nutrition, and goal-attainment data; report is rendered on-screen and made available for export (FR-15).

UC-7: Log Meals and Nutrition

- **Actors:** User
- **Main Flow:** User selects or searches for a food item; system records the meal with estimated calories and macronutrients against the daily nutrition log.

UC-8: Manage Reminders and Notifications

- **Actors:** User, Nutritionist/Trainer, System
- **Main Flow:** System monitors defined trigger conditions (scheduled workout, hydration interval, meal time, missed target); when a condition is met, a notification is pushed to the relevant user via app/SMS/email.

UC-9: Manage User Profile

- **Actors:** User
- **Main Flow:** User creates, edits, or updates their profile, including age, weight, height, fitness goals, and dietary preferences.

UC-10: Configure System Settings

- **Actors:** System Administrator
- **Main Flow:** Administrator manages user roles, recommendation/alert thresholds (FR-19), and integration credentials for external APIs.

8. Use Case Diagram

User	Track Daily Activity	Generate Workout Plan	Wearable Device System
	Generate Diet Plan	Sync Wearable Data	
Nutritionist / Trainer	View Progress Dashboard	Generate Progress Reports	AI Recommendation Engine
	Log Meals & Nutrition	Manage Reminders & Notifications	
System Administrator	Manage User Profile	Configure System Settings	



Primary actor



External system (secondary actor)

9. DATA REQUIREMENTS

Data Entity	Key Attributes
User Profile	User ID, name, age, weight, height, BMI, fitness goal, dietary preferences, health conditions
Activity Log	Timestamp, user ID, steps, calories burned, workout type, duration, heart rate
Nutrition Log	Timestamp, user ID, meal type, food item, calories, macronutrients (carbs/protein/fat)
Workout Plan	Plan ID, user ID, exercises, sets/ reps or duration, schedule, status
Diet Plan	Plan ID, user ID, meal recommendations, target calories, macronutrient targets, status
Wearable Device Data	Device ID, user ID, synced metric type, value, timestamp
Reminder / Notification	Notification ID, type, severity, target user, timestamp, status (read/unread)
Progress Report	Report ID, user ID, date range, metrics summary, generated timestamp

Data shall be stored in a structured cloud database with time-series support for activity and nutrition logs, and shall support historical querying for trend analysis and recommendation-model retraining. Activity and wearable telemetry are expected to arrive at high frequency (per-user updates every few seconds to minutes), so the database layer should support time-series-optimized storage (e.g., partitioned tables or a dedicated time-series database) separate from relational entities such as User Profile and Workout/Diet Plan.

10. EXTERNAL INTERFACE REQUIREMENTS

User Interfaces: Web-based dashboard for users and trainers/nutritionists (desktop); mobile app for users with activity logging, meal logging, and reminder notifications.

Hardware Interfaces: Consumer wearable devices (fitness bands, smartwatches) capturing steps, heart rate, and sleep data, communicating over Bluetooth/Wi-Fi to a companion app which relays data to the cloud backend.

Software Interfaces:

- REST API integration with third-party wearable-device platforms (e.g., Fitbit, Google Fit, Apple Health) for activity and biometric data
- REST API integration with a nutrition/food database provider for calorie and macronutrient lookup
- AI recommendation model-serving API (internal microservice) for workout and diet personalization
- Email/SMS/push notification gateway for reminders and alerts

Communication Interfaces: Encrypted HTTPS/TLS communication between wearable companion apps, mobile apps, and the cloud backend; lightweight REST/JSON messaging for periodic wearable data synchronization.

11. SYSTEM FEATURES

- **Real-Time Activity & Nutrition Tracking** – Live logging of physical activity, calories burned, and dietary intake for every user.
- **Personalized Workout Engine** – AI-driven generation of workout schedules matched to user goals, fitness level, and progress.
- **Personalized Diet Engine** – Meal recommendations that balance calorie and macronutrient targets against user preferences and health conditions.
- **Wearable Device Integration** – Automatic synchronization of steps, heart rate, and sleep data from supported devices.
- **Progress Analytics & Reporting** – Historical and real-time reports on activity, nutrition, and goal attainment.
- **Reminders & Notifications** – Timely alerts for workouts, hydration, and meals to encourage consistency.
- **Role-Based Access** – Tailored dashboards for users, nutritionists/trainers, and administrators.
- **Secure Health Data Management** – Encrypted storage and access control for sensitive personal health information.

12. ASSUMPTIONS AND CONSTRAINTS

Assumptions:

- Users have access to a smartphone and, optionally, a compatible wearable device.
- Wearable device providers expose an API or data feed for activity and biometric synchronization.
- Users have reasonably reliable internet connectivity for data sync and recommendation updates.
- Sufficient historical activity and nutrition data will be available to train reliable recommendation models.

Constraints:

- **Technical:** The system depends on third-party wearable-device and nutrition-database APIs; unavailability or rate-limiting of these APIs directly affects data sync and recommendation accuracy.
- **Operational:** Users must be able to review or override AI-generated workout and diet recommendations, particularly in early deployment phases, to build trust in the system.
- **Economic:** Development and third-party API/licensing costs may limit the initial nutrition database coverage to common food items.
- **Legal/Compliance:** The system must comply with regional data privacy regulations regarding the storage and processing of personal health information.
- **Time:** The personalized recommendation model requires a minimum historical data collection period (estimated 3–4 weeks per user) before reliable, individualized recommendations can be generated.

13. RISK ANALYSIS

Risk	Impact	Mitigation
Third-party API downtime (wearable/nutrition data)	Activity sync or recommendations become unavailable or stale	Implement caching and fallback to last-known-good data
Insufficient historical data for recommendation model	Inaccurate or generic recommendations	Delay personalized-model activation until minimum data threshold is met; use rule-based fallback initially
Wearable connectivity loss	Gaps in real-time activity monitoring	Flag device as offline; prompt manual entry; buffer data on-device for later sync
Health data privacy non-compliance	Legal exposure, loss of user trust	Encrypt data in transit/at rest; anonymize data where feasible; enforce role-based access

14. REQUIREMENT VALIDATION (Completeness & Consistency)

Completeness Check: All eight platform objectives from the problem statement (personalized platform, activity/nutrition tracking, AI-based recommendations, wearable integration, progress visualization, reminders, secure data storage, and healthy-habit promotion) are traceable to specific functional requirements (FR-1 through FR-20) and use cases (UC-1 through UC-10), confirming stakeholder requirements have been addressed.

Consistency Check: No conflicting requirements were identified. Workout-plan generation (FR-4) and diet-plan generation (FR-5) are designed as independent but data-sharing modules, avoiding overlap. Alert generation (FR-9) is consistently triggered from the recommendation engine (FR-4/FR-5), the wearable sync module (FR-6), and the goal-tracking module (FR-18), ensuring uniform notification behavior across the system.

Ambiguity/Redundancy Review: Terms such as "real-time" and "personalized" were quantified with specific thresholds (5-second update latency, minimum 85% recommendation-model accuracy) in the Non-Functional Requirements section to remove ambiguity. No redundant requirements were found across the functional list.

Suggested Improvements:

- Add a feedback-loop requirement where actual post-workout and post-meal outcomes are compared against predicted results to continuously improve the recommendation model.
- Define a formal SLA requirement for third-party wearable-device API response times, since recommendation accuracy directly depends on this external data.
- Add an explicit data-retention and anonymization requirement to strengthen compliance readiness before final design sign-off.

15. GLOSSARY

- **BMI (Body Mass Index)** – A value derived from a person's weight and height, used as an indicator of body fatness.
- **Macronutrients** – The carbohydrate, protein, and fat components of a food or diet.
- **Personalized recommendation** – A workout or diet suggestion generated based on an individual user's profile, goals, and progress rather than a generic template.
- **Wearable synchronization** – The automated transfer of activity and biometric data from a wearable device to the platform.

16. CONCLUSION

The Fitness and Diet Tracker with Personalized Recommendations translates the fragmented, logging-only approach of conventional fitness apps into a unified, personalized, and progress-driven platform. By combining user health profiles, wearable-device telemetry, AI-based workout and diet personalization, and progress visualization, the system directly addresses the operational pain points described in the problem statement — inconsistent routines, poorly matched

diet plans, and low adherence to fitness goals — while supporting the broader goal of sustainable, healthy-lifestyle management. Beyond solving these immediate challenges, the platform establishes a foundation that users, trainers, and nutritionists can build on as engagement grows, since every additional logged activity, meal, and synced wearable metric strengthens the recommendation models and improves guidance across the entire user base rather than adding operational risk.

Ultimately, the Fitness and Diet Tracker is designed not as a one-time software deployment but as a long-term wellness companion for its users — one that adapts alongside each individual's changing goals and health status, respects evolving data-privacy expectations, and continuously narrows the gap between what users currently know about their habits and what they actually need to know to build a consistent, effective, and healthy lifestyle.

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